

Hengrong Du

CONTACT Fisk University Page: <https://hengrongdu.netlify.app/>
INFORMATION Department of Math & Computer Science E-mail: hdu@fisk.edu

RESEARCH Partial Differential Equations for Machine Learning and Complex Fluids
INTERESTS

ACADEMIC **Fisk University**, Nashville, TN, USA August 2025-Present
POSITIONS Assistant Professor, [Department of Mathematics and Computer Science](#)

University of California, Irvine, Irvine, CA, USA July 2024-June 2025
Visiting Assistant Professor, [Department of Mathematics](#)

Vanderbilt University, Nashville, TN, USA Sep. 2021-June 2024
Postdoc Scholar (Research), [Department of Mathematics](#)
Mentor: **Dr. Gieri Simonett**

EDUCATION **Purdue University**, West Lafayette, IN Aug. 2016-Aug. 2021
Ph.D., Mathematics
Advisor: **Dr. Changyou Wang**

Beijing Normal University, Beijing, China Sep. 2013-July 2016
M.S., Computational Mathematics
Advisor: **Dr. Jiequan Li**

South China Normal University, Guangzhou, China Sep. 2009-July 2013
B.S., Mathematics and Applied Mathematics

PUBLICATIONS

Diffusion and Optimal Transport in Machine Learning

1. Haoyang Zheng, Hengrong Du, Ruqi Zhang, Guang Lin. *Exploring Non-Convex Discrete Energy Landscapes: An Efficient Langevin-Like Sampler with Replica Exchange*. Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence (AAAI-26).
2. Xinyang Liu, Hengrong Du, Wei Deng, Ruqi Zhang. *Optimal Stochastic Trace Estimation in Generative Modeling*. Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS 2025).
3. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Linear Partial Gromov–Wasserstein Embedding*. Proceedings of the 13th International Conference on Learning Representations (ICLR 2025).

4. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Partial Gromov–Wasserstein Metric*. Proceedings of the 13th International Conference on Learning Representations (ICLR 2025).
5. Haoyang Zheng, Hengrong Du, Qi Feng, Wei Deng, Guang Lin. *Constrained Exploration via Reflected Replica Exchange Stochastic Gradient Langevin Dynamics*. Proceedings of the 42nd International Conference on Machine Learning (ICML 2024).
6. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *Reflected Schrödinger Bridge for Constrained Generative Modeling*. Proceedings of the 41st Conference on Uncertainty in Artificial Intelligence (UAI 2024).
7. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *On Convergence of Approximate Schrödinger Bridge with Bounded Cost*. Proceedings of the ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

Complex Fluids

8. Hengrong Du, Chuntian Wang. *Partial regularity for the three-dimensional stochastic Ericksen–Leslie Equations*. Appl. Math. Optim. 92 (2025), no. 74, 1-47.
9. Hengrong Du, Gieri Simonett, Yuanzhen Shao. *On a thermodynamically consistent model for magnetoviscoelastic fluids in 3D*. J. Evol. Equ. 24 (2024), no. 9, 1-51.
10. Hengrong Du, Yuanzhen Shao, Gieri Simonett. *Well-posedness for magneto-viscoelastic fluids in 3D*. Nonlinear Anal. Real World Appl. 69 (2023), no. 103759.
11. Hengrong Du, Tao Huang, Changyou Wang. *Weak compactness property of simplified nematic liquid crystal flows in dimension two*. Math. Z. 302 (2022), no. 2, 2111-2130.
12. Hengrong Du, Changyou Wang. *Global weak solutions to the stochastic Ericksen–Leslie system in dimension two*. Discrete Contin. Dyn. Syst. 42 (2022), no. 5, 2175-2197.
13. Hengrong Du, Changyou Wang. *Partial regularity of a nematic liquid crystal model with kinematic transport effects*. Nonlinearity 34 (2021), no. 5, 3001-3045.
14. Hengrong Du, Yimei Li, Changyou Wang. *Weak solutions of non-isothermal nematic liquid crystal flow in dimension three*. J. Elliptic Parabol. Equ. 6 (2020), no. 1, 71-98.

15. Hengrong Du, Xianpeng Hu, Changyou Wang. *Suitable Weak Solutions for the Co-rotational Beris–Edwards System in Dimension Three*. Arch. Ration. Mech. Anal. 238 (2020), no. 2, 749–803.

Calculus of Variations

16. Hengrong Du, Nung Kwan Yip. *Stability of self-similar solutions to geometric flows*. Interfaces Free Bound. 25 (2023), no. 2, 155–191.
17. Hengrong Du, Qinfeng Li, Changyou Wang. *Compactness of M -uniform domains and optimal thermal insulation problems*. Adv. Calc. Var. 16 (2023), no. 1, 17–43.

PREPRINTS

1. Hengrong Du, Qi Feng, Changwei Tu, Xiaoyu Wang, Lingjiong Zhu. *Non-Reversible Langevin Algorithms for Constrained Sampling*. [arXiv:2501.11743](https://arxiv.org/abs/2501.11743), 35 pp.
2. Yikun Bai, Huy Tran, Hengrong Du, Xinran Liu, Soheil Kolouri. *Fused Partial Gromov-Wasserstein for Structured Objects*. [arXiv:2401.03662](https://arxiv.org/abs/2401.03662), 54 pp.

AWARDS

- Bilsland Dissertation Fellowship June 2020–May 2021
It was intended to give the most accomplished final-year PhD candidates an opportunity to complete the dissertation within the 2020–21 academic year by devoting full-time effort to research and writing.

TALKS

- Analysis Seminar March 2026
University of Oklahoma
- 45th Annual WKU Mathematics Symposium November 2025
West Kentucky University
- Southern California Applied Math Symposium April 2025
University of California, Riverside
- PDE & Applied Mathematics seminar January 2025
University of California, Riverside
- PDEs & Applications Seminar January 2025
Queen’s University
- Applied Math Seminar October 2024
The University of Alabama
- PDEs and Fluid Mechanics and Atmospheric Sciences April 2024
West Virginia University
- AMS Special Session on Fluids: Analysis, Applications, and Beyond March 2024
Florida State University

	• Financial Mathematics Seminar Florida State University	March 2024
	• AMS Special Section on Dynamics and Regularity of PDEs Joint Mathematics Meeting	January 2024
	• Seminar on Analysis and Stochastic Analysis Auburn University	November 2023
	• PDE Seminar University of Tennessee, Knoxville	October 2023
	• PDE/Applied Math Seminar Indiana University, Bloomington	October 2023
	• Richard F. Barry Jr. Seminar Series Old Dominion University	September 2023
	• The 13th AIMS Conference University of North Carolina Wilmington	June 2023
	• AMS Spring Central Sectional Meeting University of Cincinnati	April 2023
	• Undergraduate Seminar in Mathematics Vanderbilt University	March 2023
	• Analysis Seminar Wayne State University	March 2023
	• 3rd Biennial Meeting of SIAM Pacific Northwest Section Washington State University	May 2022
	• AMS Spring Central Virtual Sectional Meeting Purdue University	March 2022
	• Graduate Online Mini-course Beijing Normal University	November 2021
	• PDE Seminar Fall 2021 Vanderbilt University	September 2021
	• Analysis Seminar The University of Alabama	February 2021
	• PDE Seminar Purdue University	September 2020
	• Student Analysis Seminar Purdue University	February 2020
	• Student Analysis Seminar Purdue University	February 2020
	• PDE Seminar Purdue University	January 2020
CONFERENCE PARTICIPATION	• 2026 Rivière-Fabes Symposium on Analysis and PDE University of Minnesota, Twin Cities campus	May 1–3, 2026
	• Tennessee Quantum Materials Workshop Tennessee State University	Nov. 7, 2025
	• Lean for PDEs, an ICARM & SLMATH collaboration (Virtual)	

The Simons Laufer Mathematical Sciences Institute	Oct. 6-9, 2025
• Summer School on Theory and Applications of Elliptic PDE	
University of California, Irvine	June 23-27, 2025
• Rivière-Fabes Symposium on Analysis and PDE	
University of Minnesota	Apr. 19-21, 2023
• Shanks Workshop on Advances in Mathematical and Theoretical Biology	
Vanderbilt University	Mar. 17-19, 2023
• Workshop on Geometry and Analysis of Fluid Flows	
Stony Brook University	Jan. 16-20, 2023
• AMS Fall Central Sectional Meeting	
University of Texas at El Paso	Sep. 17-18, 2022
• Shanks Workshop on Mathematical Aspects of Fluid Dynamics	
Vanderbilt University	Feb. 19-20, 2022
• KUMUNU-ISU Conference on PDE, Dynamical Systems, and Applications	
University of Nebraska-Lincoln	Oct. 23-24, 2021
• AMS Fall Western Sectional Meeting	
Online	Oct. 23-24, 2021
• The 84th Midwest PDE Seminar	
Illinois Institute of Technology	Oct. 26-27, 2019
• Midwest Geometry Conference 2019	
Iowa State University	Sept. 6-8, 2019
• The 83rd Midwest PDE Seminar	
Indiana University Bloomington	Mar. 30-31, 2019

TEACHING EXPERIENCE	Instructor at Fisk University	Fall 2025-Present
	• NSCI 285 - Bioinformatics	Spring 2026
	• MATH 110 - Precalculus	Spring 2026
	• MATH 130 - Calculus II	Spring 2026
	• MATH 240 - Linear Algebra	Spring 2026
	• MATH 125 Discrete Mathematics	Fall 2025
	• MATH 130 Calculus II	Fall 2025
	Instructor at University of California, Irvine	Fall 2024-Spring 2025
	• MATH 3A Introduction to Linear Algebra	Spring 2025
	• MATH 112B Introduction to Partial Differential Equation	Winter 2025
	• MATH 3D Elementary Differential Equations	Fall 2024
	Instructor at Vanderbilt University	Fall 2021-Spring 2024
	• MATH 3100 Introduction to Analysis	Spring 2024
	• MATH 2400 Differential Equations with Linear Algebra	Fall 2023
	• MATH 3100 Introduction to Analysis	Spring 2023
	• MATH 3100 Introduction to Analysis	Fall 2022
	• MATH 1301 Accelerated Single-Variable Calculus II	Fall 2022
	• MATH 2420 Methods of Ordinary Differential Equations	Spring 2022
	• MATH 2400 Differential Equations with Linear Algebra	Spring 2022
	Recitation Instructor at Purdue University	Fall 2016-Summer 2021

- MA 26200 Linear Algebra And Differential Equations
- MA 16200 Plane Analytic Geometry And Calculus II
- MA 16500 Plane Analytic Geometry And Calculus I

Fall 2019
Spring 2019
Fall 2018